

WASP-2b

R-1075

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-2_151006 · created 2026-07-30T14:34:54+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457301.6336 +/- 0.0022 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.143 +/- 0.022
Transit depth (Rp/Rs)^2	2.05 +/- 0.62 [%]
Orbital Inclination (inc)	89.72 +/- 1.02
Airmass coefficient 1 (a1)	1.121 +/- 0.026
Airmass coefficient 2 (a2)	-0.095 +/- 0.053
Scatter in the residuals of the lightcurve fit is	2.34 %
Best Comparison Star	#2 - [131, 262]
Optimal Aperture	3.44
Optimal Annulus	6.34
Transit Duration (day)	0.087 +/- 0.0025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.143 $\pm$ 0.022 vs 0.1326 (deviation 0.18 σ)
Duration fit vs published	0.087 d vs 0.0737 d (deviation 2.0 σ)
Mid-time O-C	1.2 min
Depth SNR	2.5
Residual scatter	2.34 %

Light curve

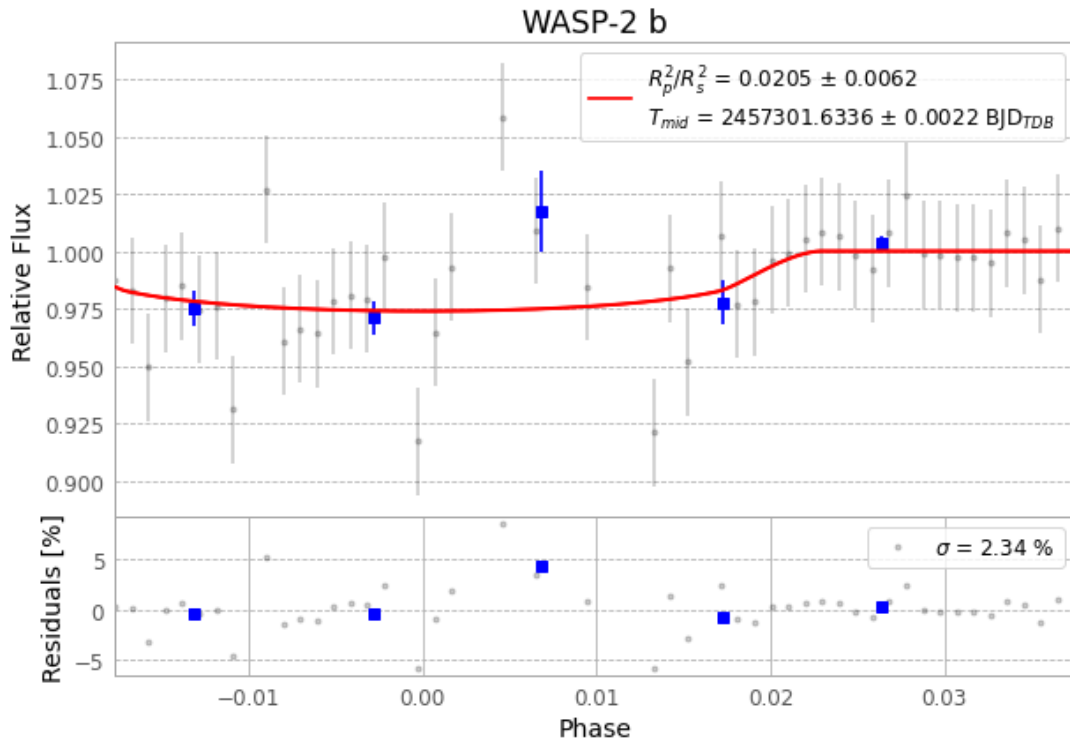


Figure 1 — FinalLightCurve_WASP-2 b_2015-10-05.png (SHA-256 622246732ac3b9b1...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [189, 279], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9a7eb2d261525be4...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

58 FITS frames (38 MB), WASP-2151006021212.FITS → WASP-2151006050312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-151006.log (d8cb649c90cf...), FinalLightCurve_WASP-2 b_2015-10-05.png (622246732ac3...), FinalLightCurve_WASP-2 b_2015-10-05.csv (947284800a12...)

Compute resources

Wall time 140.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 14:34Z.